

Airbnb Global Performance Dashboard – Comprehensive Business Intelligence Case Study

Executive Summary

This Power BI solution was developed to analyze Airbnb marketplace performance across listings, hosts, reviews, pricing, ratings, and trust metrics. The dashboard provides executive-level visibility through KPI cards, trend analysis, Pareto analysis, heatmaps, seasonality tracking, and interactive bookmark navigation. The objective is to transform raw Airbnb data into actionable business insights that support strategic decision-making.

Business Problem

Airbnb operates in a highly competitive marketplace where hosts, guests, pricing, and customer experience directly affect business performance. Stakeholders require a centralized reporting solution to monitor growth, evaluate service quality, identify top-performing cities, and understand customer engagement patterns.

Project Objectives

1. Track marketplace growth.
2. Monitor listing inventory.
3. Evaluate host ecosystem performance.
4. Measure customer satisfaction.
5. Understand review behavior.
6. Identify city-level opportunities.
7. Improve executive reporting.

Dataset Understanding

The project uses Listings and Reviews datasets. Core dimensions include City, Host, Room Type, Property Type, Review Metrics, and Rating Metrics. Together these provide a 360-degree view of Airbnb performance.

Data Modeling Approach

A star-schema style approach appears to be used. Listings acts as the primary fact table while reviews provide customer engagement metrics. Dimension attributes such as city, room type, and host information support filtering and aggregation.

KPI Framework

Total Listings, Cities, Hosts, Property Types, Reviews, Average Ratings, Trust Indicators, Review Frequency, and Market Share metrics form the KPI layer of the solution.

Overview Page Analysis

The Overview page functions as the executive landing page. KPI cards immediately communicate marketplace scale. The lifecycle trend chart shows Airbnb progressing through introduction, growth, maturity, decline, reinvention, and COVID-19 disruption phases.

Overview Visual Design Rationale

Cards were chosen because executives consume KPI information quickly. The line chart is appropriate because it visualizes changes over time. Narrative text was embedded directly into the

dashboard to transform data into business context.

Screenshot – Overview Page

Embedded dashboard screenshot.

Ratings Page Analysis

The Ratings page combines market-share analysis, pricing analysis, and customer satisfaction analysis. A Pareto-style chart identifies cities contributing the largest share of listings and reviews. The page demonstrates advanced analytical thinking beyond simple descriptive reporting.

Market Share Analysis

Paris, New York City, and Sydney account for a substantial share of marketplace activity. The cumulative percentage line enables stakeholders to identify concentration risk and understand which markets contribute the majority of performance.

Pricing Analysis

Average room price comparison reveals hotel rooms command premium pricing. Entire places represent another high-value segment. This insight can guide pricing strategy and investment decisions.

Ratings Heatmap Analysis

The heatmap compares Accuracy, Cleanliness, Communication, Location, and Value. Mexico City and Rio de Janeiro emerge as strong performers while lower-performing cities highlight potential improvement opportunities.

Bookmark Navigation Design

The star icon and magnifying-glass icon were converted into bookmark triggers. This creates an application-like user experience. Users can switch between Overall Rating and Detailed Rating views without leaving the page. This demonstrates advanced Power BI UX design skills.

Screenshot – Ratings Page

Embedded dashboard screenshot.

Reviews Page Analysis

The Reviews page focuses on customer engagement and trust. It combines review-frequency analysis, seasonal review distribution, and verification metrics to understand customer and host behavior.

Review Frequency Analysis

The Pareto chart shows that most customers leave a review only once. The cumulative percentage line reveals that a very small group contributes a disproportionately large number of reviews.

Seasonality Analysis

The monthly review-share visualization reveals seasonal travel patterns. This information can be used for demand forecasting, staffing decisions, and marketing campaigns.

Trust and Verification Analysis

Trust is critical in the sharing economy. The dashboard evaluates profile verification and profile-picture adoption. More than two-thirds of hosts appear fully verified, strengthening platform credibility.

Screenshot – Reviews Page

Embedded dashboard screenshot.

Inferred DAX Measures

Based on the visuals, likely measures include Total Listings, Total Reviews, Distinct Cities, Distinct Hosts, Average Price, Average Ratings, Superhost Listings, Monthly Review Share %, Review Frequency %, and Cumulative Percentage calculations used in Pareto analysis.

Business Insights

- Airbnb experienced strong growth before pandemic disruption.
- Listings and reviews are concentrated in a small number of cities.
- Hotel rooms achieve the highest pricing.
- Highly rated cities provide benchmarking opportunities.
- Trust indicators suggest a mature host ecosystem.

Recommendations

Expand high-performing room categories, improve underperforming cities, strengthen host education programs, use review seasonality for forecasting, and extend the dashboard with predictive analytics.

Technical Skills Demonstrated

Power BI, DAX, Data Modeling, KPI Development, Heatmaps, Pareto Analysis, Bookmark Navigation, Data Storytelling, Dashboard Design, Executive Reporting, Business Intelligence.

Portfolio Evaluation

This project demonstrates strong analytical thinking, business storytelling, and dashboard development skills. The use of narrative insights, bookmarks, Pareto analysis, and trust metrics elevates the dashboard beyond a standard reporting solution.

Conclusion

The dashboard successfully transforms Airbnb operational data into strategic insights. It balances executive KPIs, analytical depth, and user experience, making it a strong portfolio project for Data Analyst and Business Intelligence roles.

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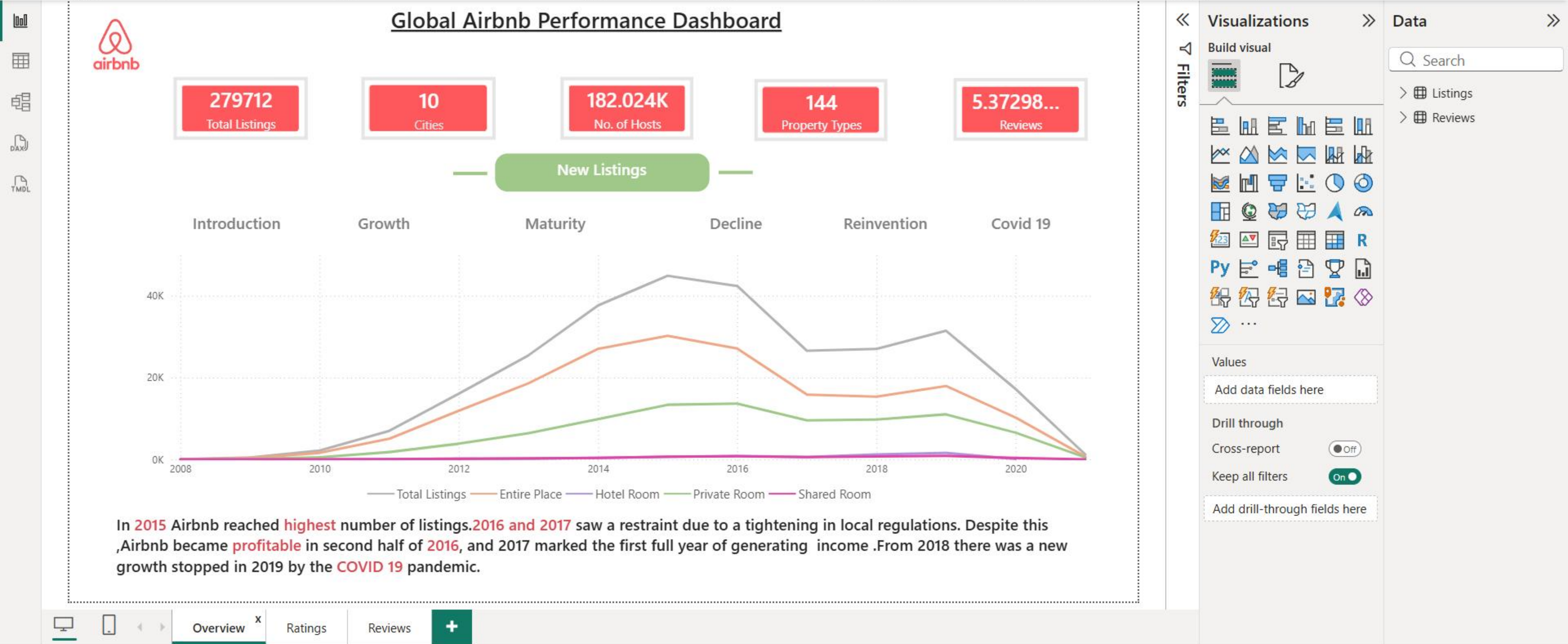
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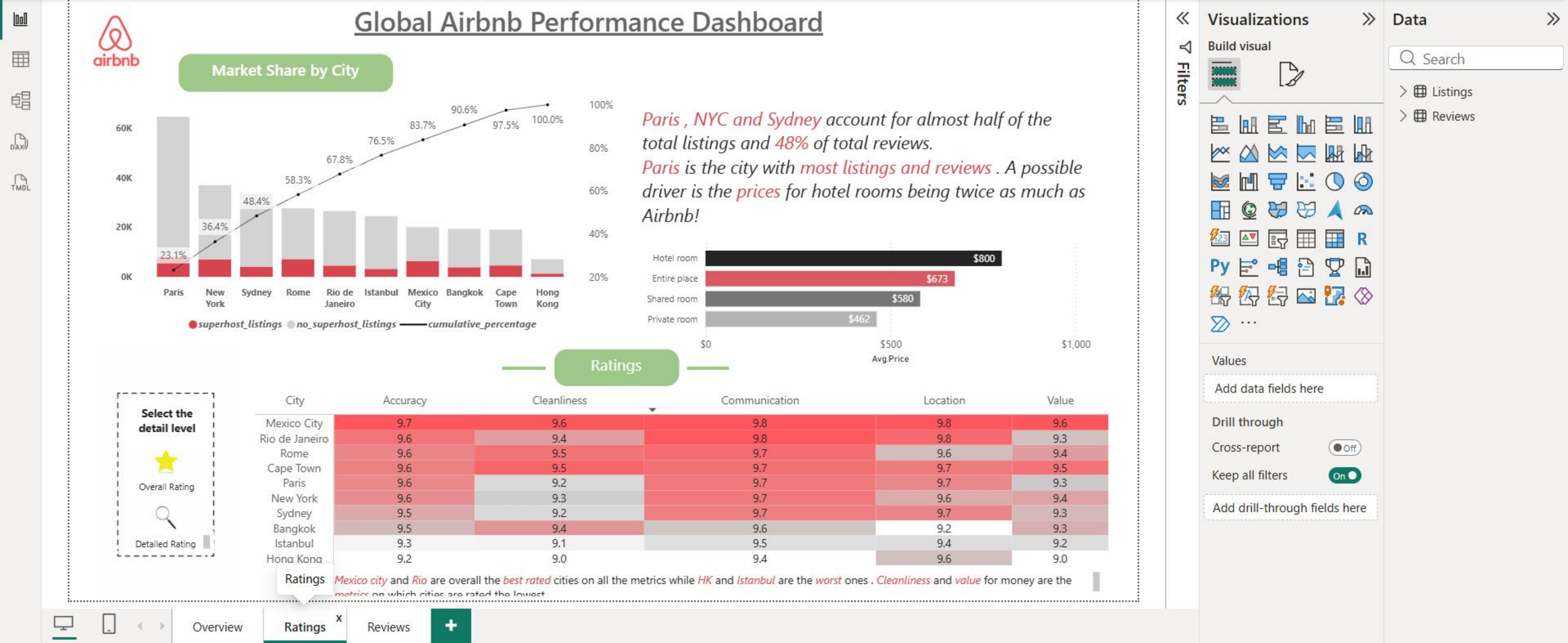
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